

The Idea of a Base

A base system groups numbers using landmark numbers that are all powers of one fixed number. Learn what a base is, how to convert between bases, and why bases make arithmetic easy.

What is a base?

The idea: A number system is **base-n** if its landmark numbers start at 1 and each next one is the previous $\times n$ — so they are all **powers of n**. You group the number using these landmark numbers.

Base 5 landmark numbers: $5^0=1$, $5^1=5$, $5^2=25$, $5^3=125$, $5^4=625$, $5^5=3125$

The Egyptian system is **base 10** (landmark numbers 1, 10, 100, 1000...). Base 10 is also called the **decimal** system. We can invent other bases too — base 5, base 4, and so on.

Converting from base 10 into another base

Group the number using the largest landmark number that fits, then the next, and so on. The count of each landmark number becomes a digit. (Quick method: keep dividing by the base and read the remainders from bottom to top.)

$$143 = (1 \times 125) + (0 \times 25) + (3 \times 5) + (3 \times 1) \rightarrow 1033 \text{ in base 5}$$

Converting back to base 10

Multiply each digit by its landmark number (its place value) and add.

$$1033 \text{ in base 5} = 1 \times 125 + 0 \times 25 + 3 \times 5 + 3 \times 1 = 143$$

Why bases are powerful: the product of any two landmark numbers is another landmark number (because powers add: $10^2 \times 10^3 = 10^5$). This makes multiplication far easier than in the Roman system, where landmark sizes are irregular.

Prerequisite refresher: A digit's *place value* is the landmark number it sits over. In base 5 the places are ..., 125, 25, 5, 1 (from left to right).

Where marks slip away

Trap 1 — Digits must be less than the base. In base 5 the only digits are 0, 1, 2, 3, 4. You never write a "5" in base 5 — five of one landmark make the next.

Trap 2 — Don't forget the zeros. If a landmark number isn't used, write 0 in that place (143 → 1033 in base 5, not 133).

Trap 3 — Place value vs digit. When converting back, multiply each digit by its landmark number, not by the base each time blindly.

Trap 4 — Largest landmark first. Always start grouping from the biggest landmark number that fits.

Slip 5 — Show the base (e.g. write "1033 in base 5") so it isn't read as an ordinary number.

Model answers — write them like this

Example A. Express 143 in base 5.

Step 1: Largest base-5 landmark below 143 is 125. $143 \div 125 = 1$ remainder 18.

Step 2: $18 \div 25 = 0$ (25 doesn't fit) → digit 0. $18 \div 5 = 3$ remainder 3 → digit 3.

Step 3: Remaining 3 ones → digit 3. Reading the digits: 1, 0, 3, 3.

$$\therefore 143 = 1033 \text{ in base 5}$$

Example B. Convert 1033 (base 5) to base 10.

Step 1: Place values are 125, 25, 5, 1.

Step 2: $1 \times 125 + 0 \times 25 + 3 \times 5 + 3 \times 1 = 125 + 0 + 15 + 3$.

$$\therefore 1033 \text{ (base 5)} = 143$$

Example C. Express 16 in base 4.

Step 1: Base-4 landmark numbers: 1, 4, 16, 64. The largest that fits 16 is 16 itself.

Step 2: $16 = 1 \times 16 + 0 \times 4 + 0 \times 1 \rightarrow$ digits 1, 0, 0.

$$\therefore 16 = 100 \text{ in base 4}$$

Example D. Why is multiplication easy in a base system?

Step 1: Every landmark number is a power of the base.

Step 2: Multiplying two landmark numbers just adds the powers, e.g. $10^2 \times 10^3 = 10^5$ — another landmark number.

$\therefore$ Products of landmark numbers stay landmark numbers, so multiplication is simple

Next: [try the worksheet](#), then [check the answer key](#).